THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

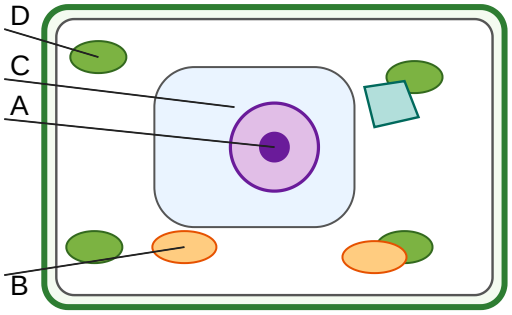
ANSWER KEY & MARKING SCHEME — SCIENCE · CLASS 9 · SET 03

Max. Marks: 80 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. In the plant cell shown, which labelled structure stores cell sap and helps to keep the cell turgid?



(a) A (b) B (c) C (d) D

Ans: (c) The large central vacuole (C) stores cell sap and maintains turgidity.

2. Which of the following cell structures is found in both prokaryotic and eukaryotic cells?

(a) Nuclear membrane (b) Mitochondria

(c) Ribosomes (d) Golgi apparatus

Ans: (c) Ribosomes are present in all cells.

3. Ropes and mats are made from the fibres of coconut husk. The tissue that gives these fibres their strength is made of:

(a) living cells with thin walls (b) living cells with thickened corners

(c) dead cells with lignified thick walls (d) cells with dense cytoplasm that divide

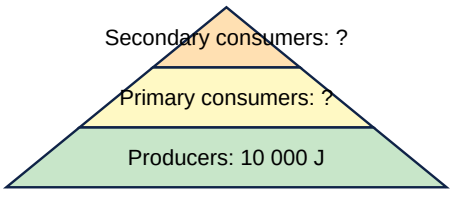
Ans: (c) Sclerenchyma – dead cells with thick lignified walls.

4. Which tissue stores fat and also acts as an insulator below the skin?

(a) Areolar tissue (b) Adipose tissue (c) Cartilage (d) Tendon

Ans: (b) Adipose tissue is the fat-storing connective tissue.

5. In the energy pyramid shown, producers fix 10 000 J of energy. Assuming the 10 % law, the energy available to the secondary consumers is:



(a) 10 J (b) 100 J (c) 1000 J (d) 10 000 J

Ans: (b) 10 000 → 1000 (primary) → 100 J (secondary).

6. Bryophytes (mosses) differ from pteridophytes (ferns) because bryophytes:

(a) have well-developed vascular tissue
(b) lack vascular tissue and grow in moist places
(c) produce seeds
(d) bear flowers

Ans: (b) Mosses lack xylem and phloem and need moisture.

7. In a flowering plant the male gamete is present in the:

(a) ovule (b) ovary (c) stigma (d) pollen grain

Ans: (d) Pollen grains carry the male gametes.

8. **Assertion (A):** Skeletal muscle is called voluntary muscle.
Reason (R): Its cells are striated and multinucleate.

Ans: (b) Both true, but 'voluntary' refers to conscious control, not the structure of fibres.

9. **Assertion (A):** The amount of carbon dioxide in the atmosphere would remain nearly constant if photosynthesis and respiration/decay were balanced.
Reason (R): Plants remove CO_2 by photosynthesis while respiration and decomposition return it.

Ans: (a) R describes the two opposite processes which keep CO_2 constant.

10. Differentiate between xylem and phloem on the basis of the material they transport and the type of cells.

Xylem: water and minerals; mostly dead cells (vessels, tracheids). [1] Phloem: food (sugars); living cells (sieve tubes, companion cells). [1]

OR

Name two permanent simple tissues of plants and write one function of each.

Parenchyma – storage/photosynthesis; collenchyma – flexibility; sclerenchyma – mechanical strength. [Any two × 1]

11. Define insolation. A solar panel of area 3 m^2 receives insolation of 600 W m^{-2} for 4 hours. Calculate the solar energy received (in kWh).

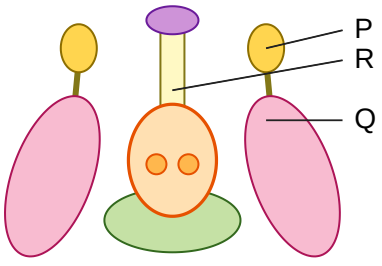
Insolation – incoming solar radiation received per unit area (per unit time) at the Earth's surface. [1] Energy = 600 × 3 × 4 = 7200 Wh [1] = 7.2 kWh [1]

OR

What is the ozone layer and where is it present? State one cause of its depletion and one harmful effect of the depletion.

Layer of ozone (O_3) in the stratosphere that absorbs harmful UV rays. [1] Cause – CFCs. [1] Effect – skin cancer/cataract/damage to crops (any one). [1]

12. The figure shows a flower with parts P, Q and R labelled. Name them and state the function of P.



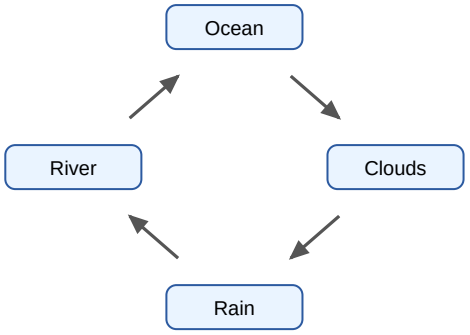
P – Anther, Q – Filament, R – Style [1½]; the anther produces pollen grains (male gametes). [1½]

13. Complete the table:

Organism	Kingdom
Paramecium	(a) _____
Bread mould (*Rhizopus*)	(b) _____
Fern	(c) _____

(a) Protista (b) Fungi (c) Plantae [1 each]

14. The figure shows the movement of water among the spheres of the Earth. Name the process by which water moves (i) from the ocean to the atmosphere, (ii) from clouds to the land and (iii) from the land back to the ocean.



(i) Evaporation [1] (ii) Precipitation (rain/snow) [1] (iii) Run-off / flow of rivers [1]

15. In a school garden, Meena observed that a young mango sapling bends in strong monsoon winds but does not break, while the trunk of an old tree is hard and rigid.

(a) Name the tissue that makes the sapling's stem flexible.

Collenchyma. [1]

(b) Why is sclerenchyma rigid?

Its dead cells have thick walls made rigid by lignin. [1]

(c) Predict what would happen to the sapling in strong winds if its stem had only sclerenchyma instead of collenchyma. Explain.

The stem would break [1]; it would be rigid and could not bend. [1]

OR

(c) Write two differences between parenchyma and sclerenchyma.

Parenchyma: living, thin-walled, stores food/photosynthesis; Sclerenchyma: dead, thick lignified walls, gives strength. [1 + 1]

16. (i) Differentiate between pollination and fertilisation.
(ii) Describe how fertilisation takes place in a flowering plant.

(i) Pollination – transfer of pollen from anther to stigma; fertilisation – fusion of male and female gametes. [2]
(ii) Pollen grain on the stigma forms a pollen tube [1] that grows through the style to the ovule [1]; male gamete fuses with the female gamete (egg) in the ovule forming a zygote. [1]

OR

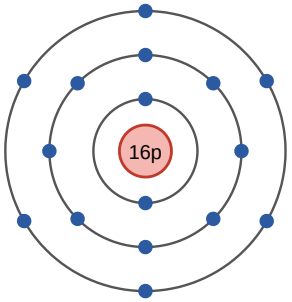
Describe any three methods of vegetative propagation, giving one example of each. State two advantages of vegetative propagation.

Cutting – rose/sugarcane; layering – jasmine; grafting – mango/apple (or budding in Bryophyllum). [3 × 1]
Advantages: plants bear fruit/flowers earlier; offspring identical to parent. [2]

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. The atom shown has 16 protons. Its valency is:



(a) 6 (b) 2 (c) 8 (d) 16

> **Ans: (b)** Configuration 2, 8, 6 – needs 2 electrons to complete the octet, so valency = 2.

18. The correct chemical formula of aluminium oxide is:

(a) AlO (b) Al_2O_3 (c) Al_3O_2 (d) AlO_3

> **Ans: (b)** Al^{3+} and O^{2-} – criss-cross gives Al_2O_3.

19. Light takes about how much time to reach the Earth from the Sun, if the distance is 1.5×10^{11} m and the speed of light is 3×10^8 m s^{-1}?

(a) 50 s (b) 500 s (c) 5000 s (d) 5 s

> **Ans: (b)** $t = 1.5\times10^{11} / 3\times10^{8} = 500$ s.

20. **Assertion (A):** The mass of a sealed flask does not change when a chemical reaction takes place inside it.
Reason (R): Mass is neither created nor destroyed in a chemical reaction.

> **Ans: (a)** R is the law of conservation of mass which explains A.

21. A student adds chalk powder to water and shines a torch through it. State one observation, name the type of mixture formed and name a method to separate the chalk.

> Light path is visible/ the mixture is turbid and chalk settles [½]; suspension [½]; filtration (or decantation) [1].

22. Write the chemical formula of (i) magnesium chloride and (ii) the chemical name of Na_2SO_4.

> (i) $MgCl_2$ [1] (ii) Sodium sulphate [1]

23. Write the number of protons, neutrons and electrons in a sodium ion, $^{23}_{11}Na^+$.

> Protons 11 [½]; neutrons 23 − 11 = 12 [1]; electrons 11 − 1 = 10 [½].

24. 40 g of a 15 % (m/m) sugar solution is diluted by adding 20 g of water. Calculate the new % m/m concentration.

> Sugar = 15 % of 40 = 6 g [1]; new mass of solution = 60 g [1]; % m/m = 6/60 × 100 = 10 % [1]

25. Name the sub-atomic particle discovered by Chadwick. State its charge, relative mass and location in the atom.

> Neutron [1]; no charge [1]; mass ≈ 1 u, present in the nucleus. [1]

OR

State any three properties of cathode rays.

Travel in straight lines; are negatively charged (deflected towards positive plate); consist of electrons; same for any gas/electrode. [Any 3 × 1]

26. The table gives the atomic number and electronic configuration of three elements X, Y and Z.

Element	Atomic number	Electronic configuration
X	11	2, 8, 1
Y	17	2, 8, 7
Z	18	2, 8, 8

(a) Which element is chemically inert?

Z (complete octet). [1]

(b) What is the valency of Y?

1 (gains one electron). [1]

(c) Write the formula of the compound formed between X and Y and state its nature.

XY [1]; ionic compound (transfer of electron from X to Y). [1]

OR

(c) Write the symbol (with charge) of the ion formed by X and state the number of protons and electrons in it.

X^+ [1]; 11 protons, 10 electrons. [1]

27. Describe an activity to show that black ink is a mixture, using paper chromatography. Write the procedure and the observation.

Materials – filter-paper strip, beaker, water (solvent), ink [1]. Draw a baseline with pencil; put a drop of ink on it [1]; dip the lower end in water without touching the spot [1]. Water rises, carrying the dyes at different rates [1]. Observation – different coloured spots/bands separate, showing ink is a mixture of dyes. [1]

OR

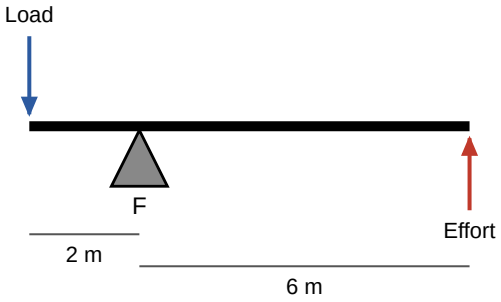
(a) Give two differences between a mixture and a compound. (b) Classify as homogeneous or heterogeneous: air, sea water, soil, brass.

(a) Mixture – components keep properties, in any ratio, separated by physical methods; compound – new properties, fixed ratio, only by chemical methods [2 × 1½ = 3 with wording] (b) Air – homogeneous; sea water – homogeneous; soil – heterogeneous; brass – homogeneous [2]

SECTION C — PHYSICS — 12 questions · 23 marks

Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

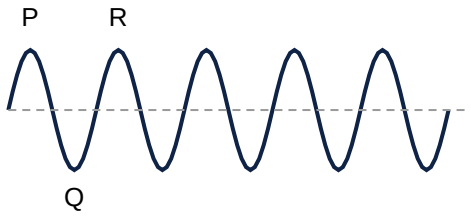
28. The lever shown has a load arm of 2 m and an effort arm of 6 m. Its mechanical advantage is:



(a) 1/3 (b) 3 (c) 4 (d) 8

Ans: (b) MA = effort arm / load arm = 6/2 = 3.

29. In the wave shown, P and R are consecutive crests and Q is a trough. The distance between P and R is the:



(a) amplitude (b) frequency (c) time period (d) wavelength

> **Ans: (d)** Distance between two consecutive crests is the wavelength.

30. A ball is thrown vertically upwards. At its highest point its velocity and acceleration are respectively:

(a) zero and 10 m s^{-2} downward (b) zero and zero

(c) 10 m s^{-1} and zero (d) 10 m s^{-1} upward and 10 m s^{-2} downward

> **Ans: (a)** At the top v = 0 but g still acts downwards.

31. Two blocks of masses 2 kg and 4 kg are pushed by the same force of 12 N on a smooth surface. The ratio of their accelerations is:

(a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 4 : 1

> **Ans: (b)** $a_1 = 6$, $a_2 = 3$ m s^{-2} → 2 : 1.

32. A machine does 3600 J of work in 1 minute. Its power is:

(a) 60 W (b) 3600 W (c) 216 000 W (d) 1 W

> **Ans: (a)** P = 3600/60 = 60 W.

33. Which of the following frequencies lies in the infrasonic range?

(a) 10 Hz (b) 2000 Hz (c) 25 kHz (d) 50 kHz

> **Ans: (a)** Infrasonic: below 20 Hz.

34. **Assertion (A):** A passenger tends to fall forward when a moving bus stops suddenly.
Reason (R): The lower part of the body stops with the bus while the upper part continues to move due to inertia of motion.

> **Ans: (a)** R correctly explains A (first law).

35. A bus increases its speed from 18 km h^{-1} to 54 km h^{-1} in 5 s. Find (i) its acceleration and (ii) the distance covered in this time.

> u = 5 m s^{-1}, v = 15 m s^{-1}. (i) a = (15−5)/5 = 2 m s^{-2} [1] (ii) s = (5+15)/2 × 5 = 50 m [1]

36. Why can sound not travel through a vacuum?

> Sound is a mechanical wave needing a material medium [1]; in vacuum there are no particles to be compressed and rarefied. [1]

OR

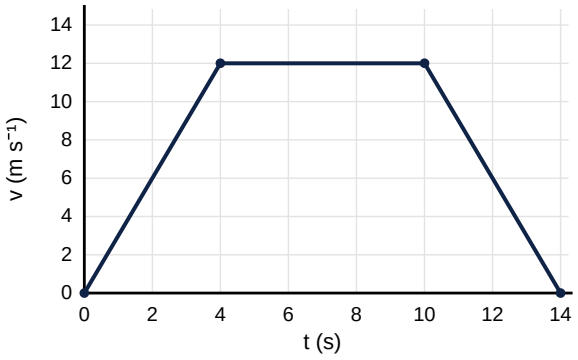
Differentiate between loudness and pitch of a sound.

> Loudness depends on amplitude; pitch depends on frequency. [1 + 1]

37. A 2 kg ball is released from a height of 20 m (g = 10 m s^{-2}). Find (i) its potential energy at the top, (ii) its kinetic energy just before it hits the ground and (iii) its speed at that instant.

> (i) mgh = 2×10×20 = 400 J [1] (ii) KE = 400 J (energy conserved) [1] (iii) ½×2×v^2 = 400 → v = 20 m s^{-1} [1]

38. Asha recorded the velocity of her scooter on a straight road. The velocity–time graph is shown.



(a) Describe the motion of the scooter between t = 4 s and t = 10 s.

> Uniform motion – constant velocity of 12 m s⁻¹, zero acceleration. [1]

(b) Find the acceleration during the first 4 s.

> a = (12 − 0)/4 = 3 m s⁻². [1]

(c) Calculate the retardation during the last 4 s.

> Retardation = 12/4 = 3 m s⁻² [2: 1 for formula, 1 for value with unit]

OR

(c) Find the total distance travelled and the average speed during the 14 s.

> Area = ½×4×12 + 6×12 + ½×4×12 = 24 + 72 + 24 = 120 m [1]; average speed = 120/14 ≈ 8.6 m s⁻¹ [1]

39. A bullet of mass 20 g moving at 250 m s^{-1} strikes a wooden block and stops after penetrating 0.1 m. Calculate (i) its initial kinetic energy, (ii) the work done by the resistive force, (iii) the average resistive force, (iv) the retardation and (v) the time taken to stop.

> (i) ½×0.02×250² = 625 J [1] (ii) 625 J [1] (iii) F = W/s = 625/0.1 = 6250 N [1] (iv) a = F/m = 6250/0.02 = 3.125 × 10⁵ m s⁻² [1] (v) t = v/a = 250/312500 = 8 × 10⁻⁴ s [1]

OR

A 50 kg boy climbs a staircase of 20 steps, each 15 cm high, in 10 s. Calculate (i) the height climbed, (ii) the work done against gravity, (iii) his power, (iv) the gain in potential energy and (v) his power if he did the same in 5 s.

> (i) 20 × 0.15 = 3 m [1] (ii) mgh = 50×10×3 = 1500 J [1] (iii) 1500/10 = 150 W [1] (iv) 1500 J [1] (v) 1500/5 = 300 W [1]